

PARSA NEDAEI

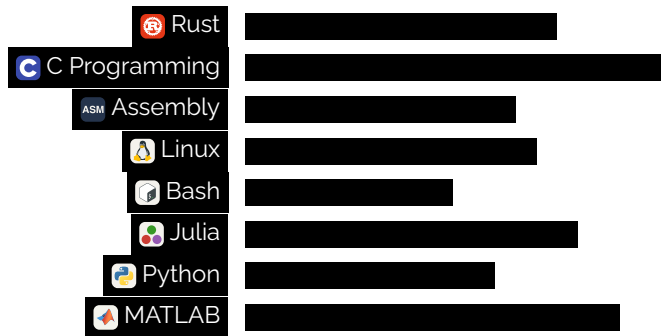
Electrical Engineer

Tehran, IR
+98 935 305 4550
parsanedaei.2001@gmail.com

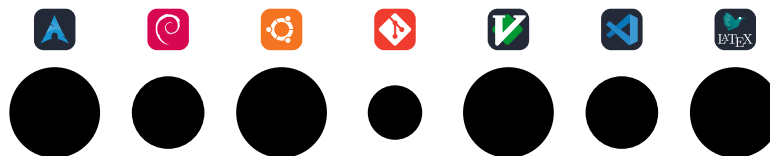
p-nedaei
parsa-nedaei

WHO AM I?

Electrical Engineering graduate and MSc student of Photonics Engineering with specialization in Optoelectronics and Semiconductors, with practical experience in electronic circuit design and hardware programming. My goal is to obtain a research or industrial position that will allow me to utilize my multidisciplinary knowledge in the areas of semiconductor device design, operating system development, and quantum computing. I am eager to contribute to the advancement of advanced technologies and knowledge transfer to the next generation in an innovative and academic environment.



TECHS & TOOLS STACK



EDUCATION

- 2024 – Present **Master's Degree - Photonics Engineering** Shahid Beheshti University
Designing, simulating, and building Thin-Film transistors for optimization in transistor parameters. Research and learning quantum computing and information theory.
- 2020 – 2024 **Bachelor's Degree - Electrical Engineering** University of Isfahan
Studied *Electrical Engineering* with major in *Electronics* and *Telecommunications*, with focus on both analog and digital aspects such as **Information Theory and Process**, **RF Circuit Design and Analysis**, **Fiber Optics**, and **Antenna Design**.

EXPERIENCE

- Oct 2022 – Jan 2023 **Teacher Assistant - Signals & Systems** Faculty of Electrical Engineering, University of Isfahan
Lecturer: Dr. Negin Sayyaf
MATLAB / Mathematics / Python
- Oct 2023 – Jan 2024 **Teacher Assistant - Engineering Electromagnetics** Faculty of Electrical Engineering, University of Isfahan
Lecturer: Dr. Hassan Zamani
COMSOL / MATLAB
- Feb 2024 – May 2024 **Teacher Assistant - Signals & Systems** Faculty of Electrical Engineering, University of Isfahan
Lecturer: Dr. Farzad Parvaresh
MATLAB / Mathematics / Python
- Feb 2024 – May 2024 **Laboratory Assistant - Laboratory of RF Circuits** Faculty of Electrical Engineering, University of Isfahan
Principal Investigator: Dr. Mohsen Mivehchi
Altium Desing / RF Electronics / ADS

June 2024 –
Aug 2024

R&D - **Fiber Laser Tuning Techniques**

Faculty of Electrical Engineering, University of Isfahan

Principal Investigator: Dr. Hamidreza Karimi Alavijeh

Lumerical Interconnect / COMSOL / Spectrum Analysis / Photonics / Laser
Physics

INTERESTS

- Semiconductor Devices & Material Science
- Quantum Computing & Information Theory
- Operating System Architectures & Embedded Systems
- Kernel Development

CURRENT RESEARCHS, STUDIES, & PROJECTS

1. Designing, simulating, and building Thin-Film transistors for optimization in transistor parameters
2. Embedded Systems using Rust
3. Research and learning quantum computing and information theory
4. Research and learning Operating System and kernel development

PROJECTS

- Simulation of Hermite-Gaussian and Laguerre-Gaussian laser beam profile with MATLAB using Nvidia CUDA
- Optical Transmission of 4x4 keypad to LCD with laser using two ATmega32 microcontrollers at transmitter and receiver side without involving TX and RX ports
- Design, simulation and building RF receiver for aviation band (This project was preparation for new major of study in University of Isfahan, Avionics)
- Design, simulation and building optical circuit for tunable rare-earth doped fiber laser with new tuning technique
- Simulation of transmission and receiving of audio signals using OFDM with MATLAB

LANGUAGES

Persian - native

English - proficient

German - Elementary

HOBBIES

Just making small projects, like Rust programs, C programming language, embedded systems, and quantum computing and information theory. Also, interested in philosophy of science and analytic philosophy.

CERTIFIED BADGES

